SHANGHAI JIAO TONG UNIVERSITY | SJTU GC

Envoix: A Hybrid Adaptive Platform for Secure End-to-End File Transfer

VE441 · Draft Story Map and Initial Engine Architecture

Team NUUB

June 16, 2026

Envoix — Elevator Pitch

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Never waste time choosing how to send a file again — Envoix pairs devices securely and automatically selects the fastest viable transfer path across LAN, direct Internet, peer-to-peer, relay, and encrypted store-and-forward conditions.

- **Zero decision cost** — no manual choice between local transfer, cloud upload, relay, or command-line networking
- **Secure adaptive transfer** — pairing, route selection, chunked resume, and BLAKE3 verification are engine-level behavior
- **Cross-platform core** — shared Rust semantics first, then Android and Apple bindings

Team Members

Team: NUUB

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Name	jAccount / uniqname	Task assignment
Sun Qizhen	qz_sun	Product architecture, adaptive routing engine, network path scoring.
Zhang Jinbin	moranxuege	PM, mobile client direction, Android/iOS UX integration.
Li Yuhao	17521218310	End-to-end encryption, QR/token handshake, secure session design.
He Yumeng	y4954he	Relay server, deployment, network testing, fallback validation.

Sub-team breakdown

Core engine: transport/session security and reliable transfer. Client integration: CLI first, Android next. Relay/testing: fallback deployment and constrained-network validation.

Draft Pack Coverage

Assignment requirement	Status	Deck location
Title page and elevator pitch	Included	Cover and slide 2, without presenter names.
Team name, members, uniq-names, tasks	Included	Slide 3.
Story map with UX stages, Skeletal, MVP, Stretch	Included	Slides 6–12.
Functional implementation breakdown, no UI widgets	Included	Story map cells describe engine/controller/-model behavior only.
Core technical barrier features marked	Included	**[CORE]** labels in the story map.
Two draft acceptance criteria only	Included	One Skeletal and one MVP criterion on slides 13–14.
Engine architecture diagram, flows, libraries/tools	Included	Slides 15–18, using git-restored architecture SVGs.

Core Value Proposition and UX Flow

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Core value proposition

For multi-device users who frequently move files across operating systems, Envoix automatically establishes a secure file-transfer session and selects the fastest available transfer path without accounts or manual tool choice.

Story-map scope

The map starts after Envoix is opened. It excludes view-level widgets and focuses on controller/model behavior and engine interfaces.

1. Pair two devices without an account.
2. Detect available transfer paths.
3. Select and establish the best viable path.
4. Transfer file data reliably and securely.
5. Recover from interruption and verify completion.
6. Preserve the same semantics across desktop and mobile.

Story Map — Stage 1: Pair Devices

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Skeletal Product	MVP	Stretch Goals
[CORE] SPAKE2 shared-token auth before metadata. QR invite encode/decode: token, candidate address, expiry, version, flags. Reject expired, malformed, weak-token, or incompatible invites.	Mobile QR scan or deep-link ingestion. Session state machine: invite received → authenticated → ready. Support QR invite and manually typed token.	Multi-device and multi-recipient invites. Self-hosted rendezvous invite payloads. Stronger audited PAKE and formal key-confirmation review.

Story Map — Stage 2: Detect Paths

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Skeletal Product	MVP	Stretch Goals
[CORE] Parse manual address or QR candidate. Discover LAN/direct candidates. Validate IPv4/IPv6 socket format. Check relay reachability; emit engine errors.	**[CORE]** Detect LAN, IPv4 direct, IPv6 direct. Probe UDP/QUIC usability. Check NAT/rendezvous feasibility, relay reachability, and encrypted store-forward availability.	NAT and rendezvous troubleshooting. DPI/throttling detection from QUIC failures and latency patterns. Estimate quality from latency, loss, throughput.

Story Map — Stage 3: Select Path

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Skeletal Product	MVP	Stretch Goals
[CORE] Baseline order: LAN/direct QUIC first, then relay fallback. Keep strategy behind transport abstraction. Support configurable relay endpoint.	**[CORE]** Automatic path order: LAN/direct QUIC → IPv6 direct → P2P hole punching → relay → encrypted store-forward. Emit selected-strategy and failure-reason events.	Obfuscated TCP/WebSocket fallback for restricted networks. Race candidates in parallel. Choose by observed speed and stability.

Story Map — Stage 4: Establish Secure Transport

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Skeletal Product	MVP	Stretch Goals
Use abstract `TransportDialer`, `TransportListener`, and `FrameConnection`. Open QUIC connection. Export channel-binding material for authentication.	**[CORE]** End-to-end file encryption above transport. Relay and store-forward paths never read plaintext. Bind session keys to pairing transcript and channel identity.	Transport obfuscation when UDP/QUIC is blocked or throttled. Additional fallback transports for restricted network environments.

Story Map — Stage 5: Transfer File Data

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Skeletal Product	MVP	Stretch Goals
[CORE] One-file typed frame protocol: `Hello`, `Ready`, `FileHeader`, `ResumeStatus`, `Chunk`, `Complete`, `CompleteAck`. Sequential chunks, configurable size, lifecycle events.	Android sender/receiver through shared Rust core. Consistent progress events across CLI and mobile clients. Same transfer semantics on both platforms.	Folder transfer through manifests. Multi-file transfer sessions. Multiple simultaneous transfer sessions.

Story Map — Stage 6: Resume and Verify

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Skeletal Product	MVP	Stretch Goals
[CORE] Sequential chunks and deterministic `.part` temp file. JSON sidecar: transfer id, file name, file size, chunk size, bytes received, next chunk index. Validate compatible state; resume from offset. BLAKE3 final hash before rename.	Per-chunk integrity verification. Preserve resumability across path changes. Examples: direct → relay, direct → server fallback.	User-controlled pause/resume. Adaptive chunk sizing. Parallel chunks and out-of-order recovery.

Story Map — Stage 7: Cross-Platform Delivery

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Skeletal Product	MVP	Stretch Goals
Desktop CLI on PC platforms. Use shared Rust core. Application layers depend on `envoix-client`, not low-level protocol crates.	Android client through UniFFI or equivalent Rust binding. Preserve identical pairing, transfer, resume, and verification semantics.	iOS/macOS native client through Swift binding. Release packaging for Windows, macOS, Linux, Android, and iOS.

Core technical barriers

Adaptive path selection; secure pairing/session authentication; reliable stateful transfer; shared cross-platform core; relay/rendezvous/store-forward fallback.

Acceptance Criterion 1 — Skeletal Product Feature

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Authenticated stateful direct QUIC file transfer

Given two desktop Envoix clients using the same valid pairing token, **when** the receiver listens on a reachable IPv4 or IPv6 address and the sender requests one file transfer, **then** the session authenticates before file metadata is sent, transfers the file in sequential chunks, persists resumable receiver state in a `.part` file and JSON sidecar after interruption, resumes from the recorded byte offset on restart, verifies the final BLAKE3 hash, writes the final file only after verification, and returns a successful transfer summary with the correct file name and byte count.

Done means

Expired, malformed, weak-token, or version-mismatched invites are rejected; no unauthenticated metadata is sent; no partial or corrupted output is silently promoted to the final file.

Acceptance Criterion 2 — MVP Feature

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Automatic multi-path selection

Given sender and receiver clients with a valid pairing invite and at least one viable network path, **when** automatic connection setup starts, **then** Envoix probes supported strategies, attempts LAN/direct QUIC, IPv6 direct QUIC, Internet P2P/hole punching, relay fallback, and encrypted server store-and-forward according to priority and availability, emits the selected strategy as a client event, keeps relay/server paths ciphertext-only, and either completes the transfer or returns a clear no-path-available error without creating a corrupted final file.

Done means

The user never chooses a network path manually. If the chosen path fails, the engine tries the next viable path and resumes from compatible chunk state instead of restarting from zero.

Engine Architecture — Client Crate Dependencies

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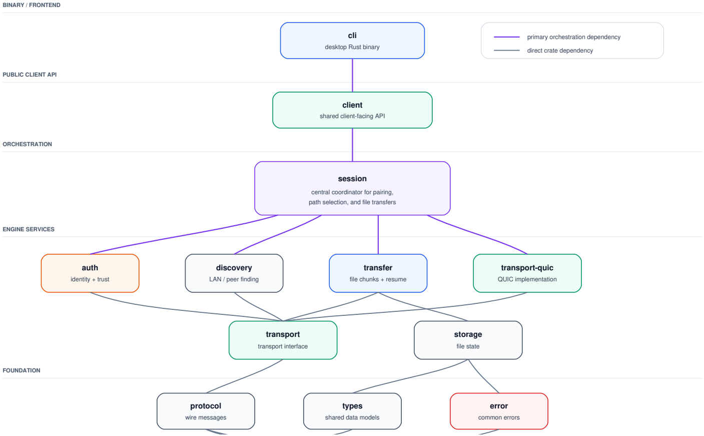

Engine Architecture — Runtime Logic Flow

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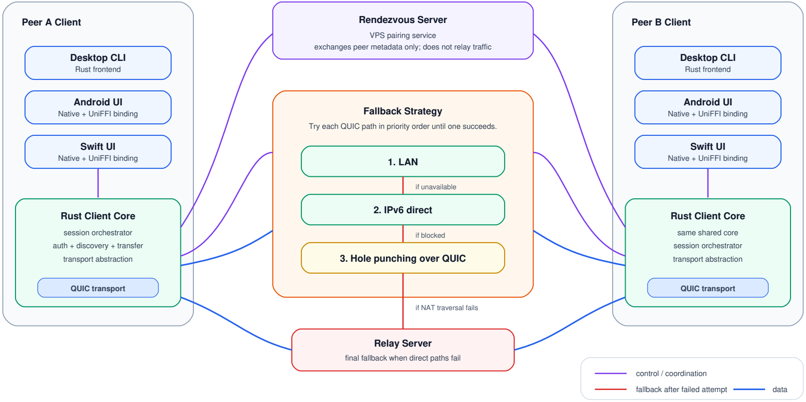

Engine Components and Planned Tools

Component	Role in engine	Key tools / libraries
`envoix-client`	Public facade for send/receive requests, config validation, progress, selected-path, auth, and failure events.	Rust, typed client API
`envoix-session`	Orchestrates pairing, candidate selection, QUIC/relay setup, and transfer engine wiring.	Tokio async runtime
`envoix-auth`	SPAKE2 token pairing, confirmation MAC, channel binding, session-key derivation path.	SPAKE2, MAC, future audited PAKE review
`envoix-qr`	Invite serialization, expiry/version validation, token generation, QR rendering.	QR payload encoding/decoding
`envoix-transport`	Abstract connection traits so transfer logic is independent of QUIC, P2P, relay, or store-forward paths.	Trait-based Rust abstraction
`envoix-transport-quic`	Current concrete direct/LAN transport.	QUIC, TLS channel material
`envoix-transfer`	Protocol state machine, typed frames, chunk streaming, complete acknowledgements, events.	Length-prefixed JSON frame protocol
`envoix-storage`	`.part` output, resume sidecar, compatible-state validation, verified rename.	JSON sidecar, BLAKE3 hash
Relay / rendezvous / server	Candidate exchange, hole-punch metadata, relay fallback, encrypted store-and-forward.	Self-hostable relay, ciphertext-only forwarding

Data and Control Flow Summary

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Control flow

1. Client receives send/receive request.
2. Session validates invite and token policy.
3. Auth completes SPAKE2 and confirmation MAC.
4. Strategy layer probes and ranks candidates.
5. Session opens selected transport or fallback.
6. Events report selected path, progress, failure reason, and completion.

Data flow

1. File metadata is sent only after authentication.
2. Transfer engine emits typed protocol frames.
3. Chunks move through QUIC, relay, or store-forward adapter.
4. Receiver writes `.part` data and JSON resume sidecar.
5. BLAKE3 verification gates final rename.
6. Relay and server paths carry ciphertext only in the MVP design.